CallTriage

Evidence-linked risk ranking for noisy, code-mixed Indian collection calls | Gnani.ai Internship Challenge

Problem

Micro-credit and collections teams handle thousands of calls a day over noisy 8kHz phone lines, mixing Hindi, Tamil, Kannada and English. QA teams can review only a tiny fraction, so threats, shaming of borrowers and false promises usually surface only after a complaint. The real question is not "what was said?" but **"which calls must a human hear first?"**

Solution

CallTriage ingests a call recording and returns a 0-100 risk score with timestamped, quote-linked evidence and playable audio, so a manager reviews the few calls that matter instead of sampling at random.

How it works

1 Segment	Silence-based splitting into short clips. Each clip's start time becomes the evidence timestamp (the API returns none).
2 Transcribe	Gnani.ai Prisma speech-to-text per clip, with retry and backoff for gateway timeouts.
3 Detect (hybrid)	Rule lexicon with fuzzy matching for hard violations + a local LLM (Qwen2.5 via Ollama) for context, promises and sentiment.
4 Score	Weighted violations (threat, family disclosure, abuse, false promise) combined into a 0-100 risk score.
5 Review	Streamlit dashboard: ranked queue, exact quote, audio proof, promise log, live upload.

Validation

Calls were downsampled to 8kHz and mixed with traffic noise at 20, 10 and 5 dB SNR. Key compliance phrases were still transcribed and flagged under heavy noise. The test set is small and synthetic, so this is a feasibility signal, not a final accuracy claim.

Who benefits

Lenders and compliance teams monitor agent conduct at scale. Borrowers gain protection because a threat or false promise leaves timestamped evidence.

Honest limitations and next steps

- Synthetic, self-recorded calls only; every flag needs human review, not a verdict.
- Speakers are inferred by the LLM (no diarization); Tamil and Kannada rules need validation.
- Next: speaker diarization, a labelled test set, agent scorecards, disclosure and calling-hours checks.

Code: https://github.com/ismailkhan-17h/calltriage | **Stack:** Python, Streamlit, Gnani.ai Prisma, Ollama (Qwen2.5), pydub, rapidfuzz